MALWARE ANALYSIS REPORT · TLP:CLEAR

HDDCryptor / “Mamba” Disk-Encrypting Ransomware

Static, dynamic and behavioural analysis of a Windows PE32 sample abusing the open-source DiskCryptor utility.

REPORT DATE	ANALYSIS PLATFORM	CLASSIFICATION
04 October 2026 · Rev. 2 (execution log added)	FLARE-VM (isolated)	Ransomware / Bootkit-style disk locker

SEVERITY

Critical

1. Executive Summary

> The sample is a 32-bit Windows executable that deploys a legitimate disk-encryption product (**DiskCryptor 1.1.846.118**) as a weapon. After receiving a password on the command line, it extracts embedded DiskCryptor components into `C:\DC22`, installs a kernel driver, creates a rogue administrator account and a Windows service, harvests network credentials, mounts reachable network shares, then encrypts every volume and replaces the master boot record (MBR) with a bootloader that displays a ransom note. After reboot the victim sees *“Missing operating system”* and a message demanding contact with an attacker e-mail address.

No network (C2) communication was observed in FakeNet; the malware is self-contained and its only external dependency is the attacker’s contact address in the ransom note. The password is an operator-supplied launch argument (the analyst supplied `123456` in this lab), so without it the data cannot be recovered. VirusTotal detection is 66/74, with family names *Trojan.Ransom.HDDCryptor* and *exe.trojan.dcryptor*.

Execution log: the malware’s own log confirms a two-stage run (installer, then SYSTEM service), shows the operator-supplied password (analyst-entered in this lab)

in clear text, and shows local disk encryption starting even when no network share is found.

Key impact: full-disk encryption, loss of OS bootability, possible spread to mapped network shares, and a persistent SYSTEM-level service.

2. Sample Identification

Attribute	Value
MD5	`409d80bb94645fbc4a1fa61c07806883`
SHA-256	`2ecc525177ed52c74ddaaacd47ad513450e85c01f2616bf179be5b576164bf63`
SHA-512	`a99a2f17d9fbb1da9fb993b976df63afa74317666eca46d1f04e7e6e24149547d1ac7210f673caeae9b23a900528ad6ad0a7b98780eff458d3d505029a06e9ba`
File type	PE32 executable (console), Intel 80386, MS Windows; magic `4D 5A` (MZ)
Entropy / packing	6.36497 — likely unpacked (payloads stored as plain PE resources)
Toolchain	Microsoft Visual Studio 2012, linker 11.00.61030, C/C++ (LTCG)
Manifest	`requireAdministrator` — will not run correctly unprivileged
Debug path (dropped Mount.exe)	`E:\localsave\trunk\CRP\CRP\Release\Mount.pdb`

THREAT INTELLIGENCE

Source	Result
VirusTotal	66 / 74 engines
MalwareBazaar	17 / 23
Metadefender	5 YARA rules matched
Family names	Trojan.Ransom.HDDCryptor, exe.trojan.dcryptor

3. Static Analysis

IMPORTS OF INTEREST

DLL	Functions	Meaning
ADVAPI32 (13)	CreateServiceW, OpenSCManagerW, StartServiceCtrlDispatcherW, ChangeServiceConfig2W, LogonUserW, ImpersonateLoggedOnUser, CreateProcessAsUserW, AdjustTokenPrivileges…	Service creation, credential use, token impersonation, privilege escalation
USER32 (2)	ExitWindowsEx, ShowWindow	Forced reboot; hides console window
SHELL32 (2)	ShellExecuteW/A	Launches dropped tools
SHLWAPI (2)	PathFileExistsW/A	Checks for prior infection (`netpass.txt`)
KERNEL32 (93)	FindResourceW, LoadResource, WriteFile, CreateThread, GetNativeSystemInfo, Wow64DisableWow64FsRedirection…	Resource extraction, 32/64-bit selection

EMBEDDED PAYLOADS (PE RESOURCES)

Both 32- and 64-bit variants are bundled and chosen at run time: `dcrypt.sys` (kernel driver), `dcrypt.exe` (GUI), `dccon.exe` (console), `dcinst.exe` (installer), `dcapi.dll` , `netpass.exe` and `Mount.exe` . Version strings identify the DiskCryptor components as v1.1.846.118 (diskcryptor.net). The driver imports ntoskrnl, FLTMGR and HAL APIs, including volume filter and MDL routines. The ransom bootloader text is stored in the DiskCryptor boot image (`MBCD`).

CRYPTOGRAPHIC AND ANTI-ANALYSIS INDICATORS

- Detected constants: AES (Rijndael Te/Td tables), Twofish, SHA-512, CRC-32, Noekeon; CryptAcquireContext / CryptGenRandom / CryptCreateHash used for hashing and randomness. DiskCryptor supports AES, Twofish and Serpent in XTS mode with HMAC-SHA-512 key derivation.
- **Anti-debug:** `IsDebuggerPresent` . **Anti-VM:** VMware checks, plus strings referencing *VBox* services, *Deep Freeze* and uninstall-key enumeration via `reg query` .

- Hard-coded working directory `C:\DC22` and log `C:\DC22\log_file.txt`.

NOTABLE STRINGS

```
Password not set.exit
net user /add mythbusters 123456
net localgroup administrators mythbusters /add
schtasks /create /tn DefragmentService /TR "cmd.exe /c net use >> c:\dc22\netuse.txt"
C:\DC22\netpass.exe /stab C:\DC22\netpass.txt
cmd.exe /c net use m: ... /USER: ... /PERSISTENT:YES
-boot -setmbr hd0      -encrypt ptN -p <password>
time limit passed.doing clean-up and reboot...
You are Hacked !!!! Your H.D.D Encrypted , Contact Us For Decryption Key (w889901665@y
```

4. Code Review (IDA)

1. **Argument check.** If `argc < 2` the program prints “Password not set.exit” and quits. The password in `argv[1]` is copied for later use as the encryption password.

2. **Stealth.** The console window is hidden with `ShowWindow`.

3. **Environment branch.** `sub_401BF0()` selects one of two paths (assessed as an OS/environment check; not fully confirmed). If `C:\DC22\netpass.txt` already exists, setup is skipped.

4. **Path A:** adds local admin `mythbusters` and records mapped shares with `net use`. **Path B:** creates, runs and deletes the scheduled task `DefragmentService` to execute `net use` in another security context. Both then run `netpass.exe /stab` to dump saved network credentials.

5. **Service install.** Sleeps ~10 s, tries `StartServiceCtrlDispatcherW` for “My Sample Service”. On failure it builds a command line (binary path + password), opens the SCM (`0xF003F`) and installs the auto-start service `DefragmentService`, then enables `SeShutdownPrivilege` and calls `ExitWindowsEx(6, 0x80020003)` to reboot.

6. **After reboot** the service runs as SYSTEM. `ServiceMain` registers a control handler, spawns a worker thread (`sub_4024F0`), which launches `C:\DC22\Mount.exe` via `LogonUserW` / `CreateProcessAsUserW` and starts disk encryption with `dccon.exe`.

7. **Mount.exe** reads the harvested credentials, mounts shares as drive `M:` (`net use m:`), enumerates mounted drives and encrypts them ("start encrypting share drives...").

8. **Clean-up** when the time limit expires: delete the service, kill `Mount.exe`, delete `Mount.exe`, `netpass.*`, `netuse.txt` and the `mythbusters` user, then `shutdown /f /r /t 0`.

5. Dynamic Analysis

- The sample only executes properly from an **elevated command prompt** with a password argument.
- Logs print to the console; a console window flashes and closes; the system then **reboots**.
- After reboot the system is very slow with **100% disk utilisation**, driven by the *System* process (DiskCryptor driver). Killing the DiskCryptor console process stabilises usage; if the disk is saturated the system crashes.
- On the next boot the ransom note appears and the OS is reported missing.
- **No C2 traffic** observed in FakeNet.

EXECUTION LOG ANALYSIS (`C:\DC22\LOG_FILE.TXT`)

The malware writes its own progress log. The captured log confirms the two-stage execution inferred from the code review:

Stage	Log entries	Interpretation
1. Installer (user-launched, elevated)	`installing driver...` → `installing driver successfully..` → `getting share drive information...` → `Trying to create service...` → `creating service successfully. rebooting windows...`	Drops resources, installs the DiskCryptor driver, collects share/credential data, registers the service and forces a reboot.
2. Service (SYSTEM, after reboot)	`Checking resources existence. They are OK...` → `driver installed before...` → `starting serviceMain...` → `ServiceMain: Entry` → `Performing Service Start Operations` → `Waiting for Worker Thread to complete` → `ServiceWorkerThread: Entry` → `Starting Mount app...`	The service re-validates resources, skips driver reinstall, runs ServiceMain and its worker thread, which launches `Mount.exe`.
3. Mount.exe	`mount:start...` → `pass:` `123456` → `mount:mounting share drive...` → `mount:share drive not found ...` → `mount:exit Mount...`	Mount.exe received the password, searched for a network share, found none and exited.
4. Disk encryption	`start hard drive encryption...`	Local encryption starts **even though no share was found**, so share mounting is optional to the attack.

Findings from the log:

- The password `123456` is written to the log in **clear text**. **This value was entered by the analyst** as the command-line argument so the sample would execute; it is not an attacker-chosen or malware-embedded key. It matches the argument stored in the service `ImagePath` (Section 6), showing how the supplied password propagates to the log and the service. In a real incident, the attacker’s password would appear in the same two places.
- Share encryption was **not triggered** in the lab because no share drive was available (`share drive not found`). This resolves an earlier limitation: the code path exists, but it was not exercised.
- The log ends at `start hard drive encryption...`. No “time limit passed” or clean-up entry was recorded, so the self-delete and final reboot routine was not observed in this run.
- `Checking resources existence` appears twice (once per stage), consistent with the same binary running as installer and then as service.

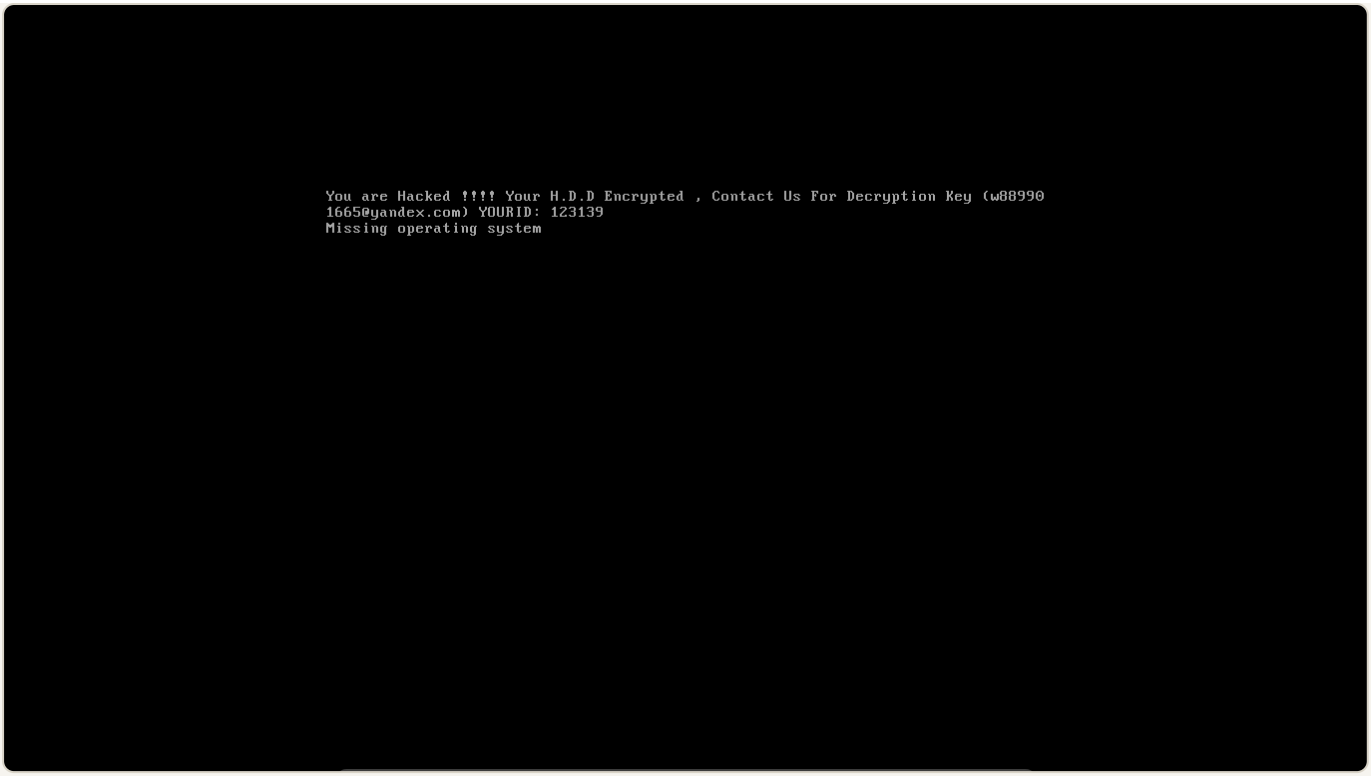


Fig. 1 — Post-encryption boot screen: ransom note and “Missing operating system”.

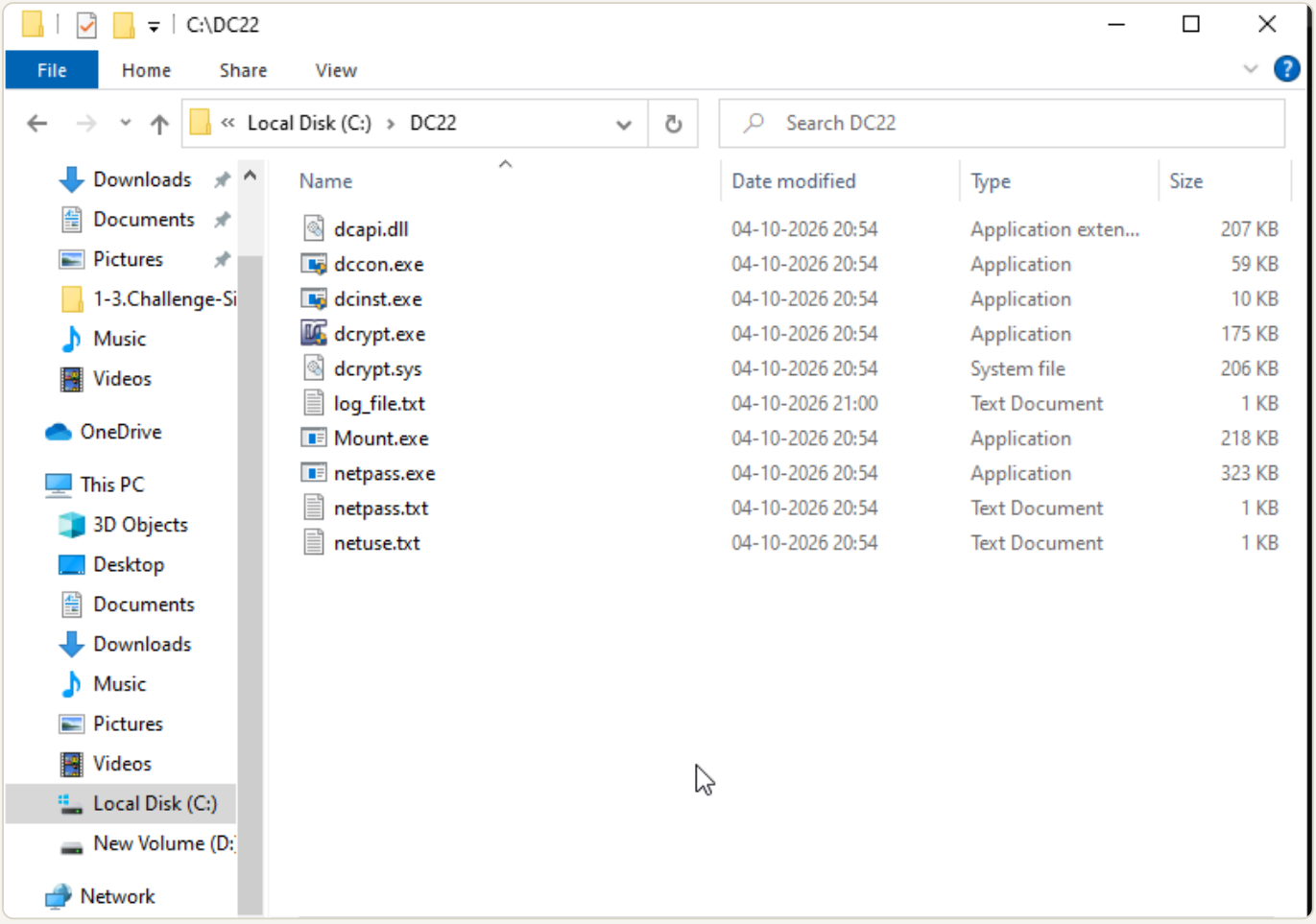


Fig. 2 — `C:\DC22` staging folder: DiskCryptor components, Mount.exe, netpass.exe/txt, netuse.txt and log_file.txt.

6. Registry and Persistence (Regshot)

Comparison window 04-Oct-2026 14:51:50 – 15:01:06 on host DESKTOP-O7TJ5FN. The raw change count is large (35,843 keys added, 104,697 values added, 549 modified) but is dominated by normal Windows/AppModel/WinGet background activity and should not be read as malicious. The security-relevant change is one new service:

```
HKLM\SYSTEM\CurrentControlSet\Services\DefragmentService
  Type        = 0x10 (Win32 own process)
  Start       = 0x2  (Automatic)
  ImagePath   = C:\Users\Analyst\Desktop\Ransomware\131.exe  123456
  ObjectName  = LocalSystem
  FailureActions configured   (mirrored under ControlSet001)
```

The name imitates the legitimate Windows `defragsvc` (which uses `svchost.exe -k defragsvc`) — a masquerading technique. No Run/RunOnce, Winlogon, IFEO, AppInit or BootExecute changes were found. Note that the service command line stores the launch password in clear text; in this lab that value (`123456`) was entered by the analyst.

7. MITRE ATT&CK Mapping

Tactic	Technique	Evidence
Execution	T1059.003 Windows Command Shell; T1569.002 Service Execution; T1053.005 Scheduled Task	`cmd /c`, schtasks `DefragmentService`
Persistence	T1543.003 Windows Service; T1136.001 Local Account	Auto-start service; `mythbusters` admin
Privilege Escalation	T1134 Access Token Manipulation	AdjustTokenPrivileges, ImpersonateLoggedOnUser
Defense Evasion	T1036.004 Masquerade Service Name; T1497 / T1622 VM and Debugger Evasion; T1070.004 File Deletion; T1027.009 Embedded Payloads	Defragment-like name; IsDebuggerPresent; clean-up routine; PE resources
Credential Access	T1555 Credentials from Password Stores	`netpass.exe /stab` (credential-recovery tool)
Discovery / Lateral Movement	T1135 Network Share Discovery; T1021.002 SMB/Admin Shares; T1078 Valid Accounts	`net use`, `net use m:`, LogonUserW
Impact	T1486 Data Encrypted for Impact; T1561.002 Disk Structure Wipe; T1542.003 Bootkit; T1529 System Shutdown/Reboot	DiskCryptor driver, MBR bootloader, ransom note, forced reboot

8. Indicators of Compromise

Type	Indicator
SHA-256	`2ecc525177ed52c74ddaaacd47ad513450e85c01f2616bf179be5b576164bf63`
MD5	`409d80bb94645fbc4a1fa61c07806883`
Directory	`C:\DC22\`
Files	dcapi.dll, dccon.exe, dcinst.exe, dcrypt.exe, dcrypt.sys, Mount.exe, netpass.exe, netpass.txt, netuse.txt, log_file.txt
Service / task	`DefragmentService`; display string “My Sample Service”
Account	Local administrator `mythbusters` (password `123456`, hard-coded in the malware; unrelated to the analyst-supplied launch argument, which happened to use the same value)
Driver / device	`dcrypt.sys`, `\Device\dcrypt`, mutex `DISKCRYPTOR_MUTEX`
Ransom contact	`w889901665@yandex.com`, ID 123139
Command lines	`dccon.exe -boot -setmbr hd0`; `dccon.exe -encrypt ptN -p …`; `net use m: …`
Log strings	`start hard drive encryption...`, `mount:share drive not found ...`, `ServiceWorkerThread: Entry`, `Starting Mount app...`
Network	None observed

9. Recommendations

DETECTION

- Alert on creation of `C:\DC22`, services named `DefragmentService`, and installation of the `dcrypt` driver outside approved use.
- Alert on `net user /add` followed by addition to Administrators, and on `schtasks` creating short-lived tasks.
- Monitor Event IDs 7045 (service install), 4720/4732 (account creation / group change), 4698 (task creation) and raw writes to `\\.\PhysicalDrive0`.
- Block or monitor DiskCryptor binaries unless sanctioned.

PREVENTION AND RESPONSE

- Enforce least privilege and application allow-listing; the sample requires admin rights.
- Restrict persistent mapped shares and stored network credentials; maintain offline, tested backups.
- Isolate affected hosts immediately. Do **not** reboot a still-running infected host; image the disk and preserve the command line / service entry, which may reveal the password used.
- Before any reboot or cleanup, collect `C:\DC22\log_file.txt` and the `DefragmentService` `ImagePath`: both can expose the encryption password, which may allow decryption with DiskCryptor.
- Recovery depends on the password or the attacker's key; avoid paying without law-enforcement and legal guidance.

10. Analyst Notes and Limitations

- Timestamps differ between tools (Regshot 14:51–15:01 vs file times 20:54–21:00 in the screenshot), likely VM clock/timezone differences; they were not reconciled.
- The purpose of `sub_401BF0` was inferred and not verified dynamically. The `netpass.exe` tool is consistent with a legitimate credential-recovery utility repurposed by the malware.
- Encryption of network shares was not observed: the execution log shows `share drive not found` in the lab, so that code path was not exercised.
- The clean-up / self-delete routine was not seen in the captured log; its trigger remains inferred from strings.

Prepared from analyst notes (FLARE-VM; HashMyFiles, DIE, CFF Explorer, HxD, Strings, IDA, Regshot, FakeNet). Handle samples only in isolated environments.